

Architectural Evaluation

Internal B2B Invoice Automation App vs. NPCI Bharat Connect

Executive Summary

As requested, I have prepared this architectural evaluation comparing the internal B2B Invoice Automation application under development with the National Payments Corporation of India's (NPCI) Bharat Connect (formerly BBPS for Business) framework. These two systems solve different parts of the B2B invoicing and settlement lifecycle and are highly complementary rather than mutually exclusive.

- **The Internal App** serves as an Accounts Payable (AP) Ingestion and Data Extraction Engine. I have designed this application to receive unstructured invoice documents (PDFs) from any supplier, read their visual layouts, validate the GST/tax calculations, and load them automatically into the ERP system (ERPNext or Tally).
- **Bharat Connect** serves as a B2B Payment and Presentation Network. It acts as a standardized national protocol that allows pre-registered ERP systems to exchange electronic invoice data and execute instant, auto-reconciled bank transfers directly between buyers and sellers.

Below is a detailed breakdown of their architectural differences, operational costs, and how they can be combined to build a complete B2B workflow.

Side-by-Side Comparison

Feature	Internal AP Automation App (This System)	NPCI Bharat Connect (BBPS)
Primary Function	Ingestion, data extraction, and mapping of incoming supplier invoices.	Electronic invoice presentment, digital payments, and automatic bank clearance.
Input Format	Any system-generated PDF (Tally, Zoho Books, SAP, custom billing apps).	Standardized XML/JSON API payloads exchanged directly between registered ERPs.
Core Technology	pdfplumber (coordinate-based spatial parser) + Gemini 2.5 Flash + custom validation scripts.	NPCI banking communication APIs and UPI/IMPS clearing networks.
GST Validation	Custom logic (compares CGST+SGST vs. IGST, extracts HSN, cross-checks vendor GSTIN state codes).	Basic data transmission (transfers whatever tax figures the seller entered; no compliance validation).
Payment Integration	None built-in (data is exported to the ERP, where payments are queued manually).	Fully integrated. Allows one-click payments (UPI, Card, Netbanking) directly from the invoice.
Reconciliation	Manual or semi-automated within the ERP post-export.	Fully automated. Matches payments to invoices instantly at the transaction level via NPCI.
Data Privacy	100% private. Data is processed in-memory and stays within the local secure database.	Public-utility model. Financial transactional data routes through NPCI's clearinghouse.

Key Operational Differences

1. Ingestion Flexibility vs. Network Onboarding

- **This App:** This is a universal solution. Because it reads the spatial layout of text, it can process invoices from 100% of suppliers immediately. The supplier does not need to change anything about how they bill; they simply send their standard PDF.
- **Bharat Connect:** This is a closed network. It only works if both the business and the suppliers are actively registered on the Bharat Connect network and use compatible ERP systems (like Zoho or Tally). It cannot read or parse standard email attachments, scanned invoices, or PDFs from unregistered suppliers.

2. Advanced GST Compliance and Validation

- **This App:** I have built custom business rules tailored for Indian GST compliance. The system automatically extracts HSN codes, cross-checks that the GST rates belong to valid slabs (12%, 18%, 28%), calculates tolerances, and compares vendor and buyer state codes to identify tax discrepancies before the data enters the books.
- **Bharat Connect:** It functions purely as a transaction delivery vehicle. It does not validate whether the invoice's tax math is correct or compliant with GST laws; it simply passes the figures from the seller to the buyer.

3. Processing Speed and Infrastructure Cost

- **This App:** By utilizing pdfplumber for local coordinate extraction and Gemini 2.5 Flash for structuring, the end-to-end processing time is under 4 seconds. The production cost is negligible - averaging about Rs. 0.15 to Rs. 0.20 per invoice when billed under pay-as-you-go pricing. Furthermore, during development and testing, I have kept the operational cost at zero (Rs. 0.00) by utilizing Google AI Studio's free tier (which offers up to 15 requests per minute and generous daily free limits) as well as Google Cloud's initial free trial credits (such as the standard Rs. 20,000 / \$300 credit for Vertex AI). There is no local server GPU overhead.
- **Bharat Connect:** Transactions are near-instantaneous. While presentment is free or low-cost, the transaction charges for B2B payments vary depending on the payment mode used (UPI, Corporate Cards, or Netbanking).

Strategic Recommendation

To build a robust, scalable B2B financial workflow, I recommend a hybrid pipeline leveraging both systems:

1. **Use the Internal App for Accounts Payable Ingestion:** For vendors who are not on the Bharat Connect network (which represents the majority of MSMEs and smaller suppliers), the spatial extraction pipeline can automatically read their PDFs, validate the taxes, and import them into the ERP.
2. **Use Bharat Connect for Payments & Reconciliation:** Once the invoices are approved and loaded into the ERP, the payment execution and bank reconciliation can be routed through the Bharat Connect B2B network.

This hybrid approach ensures 100% vendor coverage, robust tax compliance checking, and fully automated payment reconciliation.